Likelihood-ratio tests and confidence intervals

STAT 24510-30040

(Winter 2025)

1 Review: Likelihood functions

Recall: the likelihood function based on observed data X from a distribution with parameter θ is treated as a function of the parameter θ,

$$\ell(\theta) = \ell(\theta|X) = \mathbb{P}\{X = \text{observed data} \mid \text{if the true parameter value} = \theta\}$$

For example, we observe data $X = k$ (e.g., $k = 4$) under the assumption $X \sim Bin(20, p)$. The likelihood function given the data is a function of the parameter $\theta = p$.

$$\ell(\theta) = \ell(p) = \ell(p|X = k) = \binom{n}{k} p^k (1-p)^{20-k}$$

Discrete probability mass function is replaced by probability density function for continuous random variables.

For example, we observe data $X_1 = x_1, \cdots, X_n = x_n$ (e.g., $n = 3$, observed data $x_1 = 1.2, x_2 = -3, x_3 = 5.7$).

Under the assumption that X_i's are i.i.d. $\sim N(\mu, \sigma^2)$, the likelihood function given observed data is a function of the parameter pair $\theta = (\mu, \sigma^2)$.

By the independence of X_i's, the joint density function $f_{\mu,\sigma^2}(x_1, \cdots, x_n)$ of $(X_1, \cdots, X_n)$ is the product of the density functions of individual $X_i, i = 1, \cdots, n$. Given the data, the likelihood function is

$$\ell(\theta) = \ell(\mu, \sigma^2) = \ell(\mu, \sigma^2|X_1 = x_1, \cdots, X_n = x_n) = f_{\mu,\sigma^2}(x_1, \cdots, x_n) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_i-\mu)^2/2\sigma^2}$$

Log-likelihood, score function, and Fisher information

- The logarithm of the likelihood function is

  $$L(\theta) = L(\theta|X) = \log_e \ell(\theta) = \log \ell(\theta)$$

  Log-likelihood function has nice mathematical properties and often easier to work with than the likelihood function. Commonly the Euler's number e is the default base.

- The **score function** or simply the score is the derivative of the log-likelihood function with respect to the parameter.

  $$\frac{\partial}{\partial\theta} \log \ell(\theta)$$

  The score function is important in finding the maximum likelihood estimator MLE.

- The **Fisher information**

  Fisher information is defined as

  $$I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta} \log \ell(\theta|X)\right)^2\right]$$

  The follow identity (exercise)

  $$\mathbb{E}\left[\left(\frac{\partial}{\partial\theta} \log \ell(\theta|X)\right)^2\right] = -\,\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log \ell(\theta|X)\right]$$

  provides another equivalent definition

  $$I(\theta) = -\,\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log \ell(\theta|X)\right]$$

  Fisher information is an important measure of the efficiency of the MLE.

In the above, we use the notations for one-parameter case. The definitions can be generalized to parameter vectors.

Asymptotic optimality of MLE

The parameter estimator maximizing the likelihood function (the MLE) has good asymptotic properties.

Fisher's Theorem (Asymptotic distribution of MLE)

Under mild assumptions, maximum likelihood estimator $\hat{\theta}$ is asymptotically of normal distribution,

$$\sqrt{n}(\hat{\theta} - \theta) \to W \sim N\left(0, \frac{1}{I(\theta)}\right) \quad as \quad n \to \infty.$$

In other words,

$$\hat{\theta} \sim N\left(\theta, \frac{1}{nI(\theta)}\right) \quad \text{for large } n.$$

The asymptotic variance is optimal. (Ref. Casella and Berger ch. 10 for proofs.)

In practice, the $I(\theta)$ is unknown. It turn out that, replacing $I(\theta)$ by $I(\hat{\theta})$ in the variance is still valid asymptotically, that is,

$$\hat{\theta} \sim N\left(\theta, \frac{1}{nI(\hat{\theta})}\right) \quad \text{for large } n.$$

This is a more useful version of Fisher's Theorem.

2 Likelihood-ratio test (LRT)

Consider testing a null hypothesis H_0 versus an alternative H_1 (a.k.a. H_a), using the likelihood function $\ell(\theta)$. Recall the principle of maximum likelihood estimation, and define

$$\ell_0 = \text{maximum of the likelihood function when } H_0 \text{ is true,}$$
$$\ell_1 = \text{maximum of the likelihood function over all.}$$

Denote Θ the parameter space containing all possible values of θ, Θ_0 the parameter space when H_0 is true. $\Theta_0 \subset \Theta$. To compare the two hypotheses, consider the ratio of the two maxima of the likelihood functions,

$$\Lambda = \frac{\ell_o}{\ell_1} = \frac{\max_{\theta \in \Theta_0} \ell(\theta)}{\max_{\theta \in \Theta} \ell(\theta)} = LR$$

(Notes: When the parameter space is not compact, $\max$ should be replaced with $\sup$.)

Note that,

Always, $\ell_0 \leq \ell_1$.
When H_0 is true, $\ell_0 \approx \ell_1$.
When H_0 is false, $\ell_0 < \ell_1$; and $\ell_0 << \ell_1$ when H_0 is far from being true.

In other words,

$$\Lambda \begin{cases} \approx 1 & \text{if } H_0 \text{ is true;} \\ << 1 & \text{otherwise.} \end{cases}$$

It is reasonable to reject H_0 when $\Lambda \leq c$ for some critical value c.

Λ iis often called the **likelihood ratio statistic** or simply **likelihood ratio** (thus denoted as LR) even thorough it actually is the ratio of two maxima of likelihood functions, not a ratio of two likelihood functions.

Let

$$L_0 = \log \ell_0, \qquad L_1 = \log \ell_1$$

The logarithm of the ratio of the two maximum likelihood functions is $\log \Lambda = L_0 - L_1$. The statistic

$$-2\log\Lambda = -2(\log \ell_0 - \log \ell_1) = -2(L_0 - L_1)$$

is called (log) **likelihood ratio test statistic**.

Properties of likelihood-ratio test statistic

- $-2\log\Lambda \geq 0$
- $-2\log\Lambda \approx \begin{cases} near\ 0, & \text{if } H_o \text{ is true;} \\ large, & \text{otherwise.} \end{cases}$
- The likelihood-ratio statistic is asymptotically of Chi-squared distribution with degrees of freedom d (proof omitted),

  $$-2\log\Lambda = -2\log(\ell_0/\ell_1) \sim \chi^2_d \quad \text{(approximately)} \qquad \text{for large } n,$$

  where $d =$ the difference between the dimensions of the parameter spaces of Θ and Θ_0,

  $$d = \dim\{\Theta\} - \dim\{\Theta_0\}$$

Example (LRT for Binomial)

$X \sim Bin(n,p)$. If we observe $X = k$, then the likelihood and log-likelihood functions are

$$\ell(p) = \ell(p|X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\log \ell(p) = \log \binom{n}{k} + k\log(p) + (n-k)\log(1-p)$$

The MLE is $\hat{p} = k/n$.

Consider

$$\begin{cases} H_0: & p = p_0 \\ H_1: & p \neq p_0 \end{cases}$$

The null parameter space $\Theta_0 = p_0$ with dimension $= 0$,

$$\ell_0 = \ell(p_0) = \ell(p_0|X=k) = \binom{n}{k} p_0^k (1-p_0)^{n-k}$$

The full parameter space $\Theta = \{p : 0 \leq p \leq 1\}$ with dimension $= 1$,

$$\ell_1 = \max_{p\in[0,1]} \ell(p) = \ell(\hat{p}|X=k) = \binom{n}{k} \hat{p}^k (1-\hat{p})^{n-k}$$

Thus

$$LR = \Lambda = \frac{\ell_o}{\ell_1} = \frac{p_0^k (1-p_0)^{n-k}}{\hat{p}^k (1-\hat{p})^{n-k}}$$

and

$$-2\log(\Lambda) = -2\log(\ell_0/\ell_1) = -2k\log\frac{p_0}{\hat{p}} - 2(n-k)\log\frac{1-p_0}{1-\hat{p}}$$

Simplify, plug in $\hat{p} = k/n$, we obtain

$$-2\log(\Lambda) = 2k\log\frac{k}{np_0} + 2(n-k)\log\frac{n-k}{n-np_0}$$

For large n, we may use $-2\log(\Lambda) \sim \chi^2_1$ to construct test. For given test level α,

$$\text{Reject } H_0 \text{ when} \quad -2\log(\Lambda) > \chi^2_{1,1-\alpha}$$

3 Likelihood-ratio confidence intervals

Similar to the construction of the Wilson duality confidence interval (*a.k.a.* score confidence interval), the asymptotic distribution of the likelihood ratio test statistic

$$-2\log(\Lambda) \sim \chi^2_d$$

can be used to construct confidence intervals for the unknown parameter θ.

Analogous to the way Wilson confidence interval is constructed, a likelihood-ratio $100(1-\alpha)\%$ confidence interval consists of all possible values of p_0 for which the likelihood ratio test would not reject the null hypothesis at test level α.

Example (LR confidence interval for Binomial)

$X \sim Bin(10, p)$. Recall the likelihood function given observed data $X = k$ is

$$\ell(\theta) = \ell(p) = \ell(p|X=k) = \binom{n}{k} p^k (1-p)^{20-k}$$

If we observe $X = 0$, then the likelihood and log-likelihood functions are

$$\ell(p|X=0) = \binom{10}{0} p^0 (1-p)^{10} = (1-p)^{10}$$

$$\log \ell(p) = 10 \times \log(1-p)$$

The MLE for p is

$$\hat{p} = \frac{X}{n} = \frac{0}{n} = 0$$

Consider the significance test $H_0 : p = p_0$ vs $H_1 : p \neq p_0$.

The maximum likelihood and log(maximum likelihood) for H_o are

$$\ell_o = \ell(p_0) = (1-p_0)^{10}, \qquad L_0 = 10\log(1-p_0)$$

and that for H_a are

$$\ell_1 = \ell(\hat{p}) = \ell(0) = 1, \qquad L_1 = \log \ell_1 = 0.$$

The likelihood ratio test statistic is

$$-2\log(LR) = -2\log\Lambda = -2(\log \ell_0 - \log \ell_1) = -2(L_0 - L_1) = -20 \times \log(1-p_0)$$

The 95% likelihood-ratio confidence interval is the set of p_0 for which H_0 will not be rejected at test level $\alpha = 0.05$ given the observation $X = 0$. That is, the 95% likelihood-ratio confidence interval consists of all p_0 such that

$$-2\log\Lambda = -20\log(1-p_0) < \chi^2_{d=1,\alpha=0.05} = 3.84$$

From the inequalities we can obtain an interval for p_0,

$$0 < p_0 < 1 - e^{-3.84/20}$$

Therefore the 95% likelihood-ratio confidence interval for p based on the observed data $X = 0$ out of $n = 10$ is

$$\left(0,\ 1 - e^{-3.84/20}\right) = (0,\ 0.17)$$

Comparisons with difference sample sizes

If the observation $X = 0$ was out of a large sample $n = 50$ trials, then the 95% likelihood-ratio confidence interval for the true population proportion p based on the observed data (0 out of 50) would be

$$\left(0,\ 1 - e^{-3.84/100}\right) = (0,\ 0.038)$$

which is much narrower than the case $n = 10$. Naturally,

$$(0,\ 0.038) \subset (0,\ 0.17)$$

Discussions and remarks

- The above example, combined with earlier examples of Wilson confidence intervals, demonstrate the connection between hypothesis testing and construction of confidence intervals.
- There are other types of confidence intervals constructed to deal with the situations $\hat{p} \approx 0$ or $\hat{p} \approx 1$ under small sample size, when asymptotic methods are not appropriate. For example,
  - Agresti-Coull confidence interval uses the statistic $\hat{p}^* = \dfrac{k+2}{n+4}$ combining with Wald method to give a simple adjustment of the score.
  - Bayesian estimator uses test statistic $\hat{p}^* = \dfrac{k+1}{n+2}$

  The two estimators are no longer unbiased. However they provide reasonable confidence intervals when p is near 0 or 1.
- Discussion:

  Compare pros and cons of likelihood-ratio confidence intervals with corresponding Wald and score confidence intervals.
- Wald test uses the behavior of likelihood at MLE $\hat{p}$, the score test uses the likelihood at the null hypothesis p_0, while the likelihood ratio test uses both $\hat{p}$ and p_0.

Reference for C.I.: Chapter 9 in Casella and Berger; sections 7.3.3, 8.5.3, 9.3-9.5 in Rice.